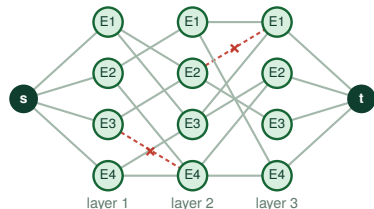


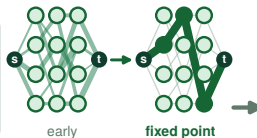
A Per-input legal expert graph



Edges are exactly the legal expert transitions.
An illegal transition (x) has no edge, so it can never be used.

B Conserved-flow dynamic (Physarum)

$$\begin{aligned} \text{conserve} \quad \mathcal{L}(w) p &= b \\ \text{flux} \quad Q &= w \odot B^T p \\ \text{reinforce} \quad D &\leftarrow D + \Delta t(|Q| - D) \end{aligned}$$



conserved at every node

one sparse path survives

The encoder sets per-edge costs; the flow relaxes to a single legal path of experts (no learned gate).

C Route and compose

Conserved flow (ours)



one legal path, 1 expert/layer: $\hat{y} = f_{E4} \circ f_{E1} \circ f_{E2}(x)$

Top-k gate (standard MoE)



The gate scores each layer independently and can chain experts in an order the task forbids.

D Train end to end: implicit differentiation

$$\begin{aligned} \text{fixed point} \quad D^* &= F(D^*, \theta) \\ \text{backward solve} \quad (I - \partial_D F)^T u &= g \\ \text{gradient} \quad \nabla_\theta &= (\partial_\theta F)^T u \end{aligned}$$

$O(E)$ memory, any depth

matrix-free VJP + CG

checked vs. finite diff.

Gradients flow through the solver without unrolling it, so training cost stays linear in edges.

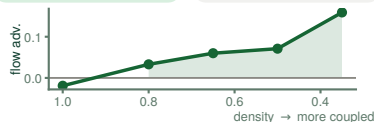
E When does it help? Coupling

coupled steps

flow > independent gate

marginalizable steps

tie (e.g. MetaQA 2-hop)



Measured: the advantage grows with coupling and crosses zero when steps are independent.

F Evaluate

- **constraint-density sweep** synthetic, 3 seeds
- **MoE FFN layer** accuracy vs. compute
- **MoE transformer** $E=6$ experts, $K=3$ layers
- **MetaQA 2-hop KGQA** 43k entities, real data

accuracy

experts / layer

illegal rate

Same encoder, same legal graph, same compute; only the router changes.